

GLA University

End Term Exam

B. Tech: II Year, 4th semester [C.S, E.C. & M.E.]

Ethics & Values (AHE: 201)

Marks: 40 Time: 2.30 Hours

There are three sections in this question paper. Section A consists of very short answer type questions, while section B & C consist of descriptive answer type questions. Attempt questions from all sections.

Section A

1) Write short notes on each of the following :(Word Limit: 50 words, Marks: 2 marks each)

a) Well-Being
b) Engineering Ethics
c) Moral Person
d) Dual logic of Experience
e) Technological Determinism

Section B

2) Answer any four of the following :(Word Limit: 200 words, Marks: 2.5 marks each)

a) Explain the concept of moral-responsibility.
b) Explain the ethics of self-Interest by giving an analysis of the computer game case.
c) What is the difference between risk-benefit analysis and cost-benefit analysis?
d) Explain the domains of knowledge.
e) What is the role of wisdom and balance in human life?

Section C

3) Answer any four of the following :(Word Limit: 300 words, Marks: 5 marks each)

a) What is philosophy of engineering?
b) How do you find technology as a means of progress?
c) Explain the dangers of technology in automobile, health and energy sectors.
d) How do rationality and critical thinking design your life in a better way?
e) Creativity enriches the meaning and value of life. Analyze the statement through examples.

SET - A

GLA University, Mathura
END-Term Examination, 2011-12
Course: B.Tech. (CSE), II-Year, IV-Sem.
Database Management Systems (ITE221)

Uni. Roll No:

Time: 2.5 Hrs. **Total Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Part - A

Attempt all. **20×.5=10**

I. In ER model, derived attribute is represented by
- Double rectangle
- Double Diamond
- Dotted ellipse
- Dotted rectangle

II. In ER model, degree of relationship is based on
- Number of attributes
- Number of entities
- Number of relations
- None of above

III. Multi-valued dependency is related to
- 1NF
- 2NF
- 3NF
- 4NF

IV. An entity set that does not have sufficient attributes to form a primary key is a
- strong entity set.
- weak entity set.
- simple entity set.
- primary entity set.

V. The statement in SQL which allows to change the definition of a table is
- Alter
- Update.
- Create.
- select.

VI. Relational Algebra is
- Data Definition Language
- Meta Language
- Procedural query Language
- Non-Procedural query Language

VII. Which of the following operation is used if we are interested in only certain columns of a table?
- PROJECTION
- SELECTION
- UNION
- JOIN

VIII. Which of the following database object does not physically exist?
- base table
- index
- view
- none of the above

IX. _____ is a restriction on access to data in a multi-user environment
- A lock
- A Transaction
- A concurrency
- None of the above

X. _____ item is called exclusive-locked, because a single transaction exclusive holds the lock on the item.
- Read -locked item
- Write-locked item
- Shared -locked item
- None of the above

XI. By data integrity we mean
- maintaining consistent data values
- integrated data values
- banning improper access to data
- not leaking data values

XII. A transaction manager is which of the following?
- Maintains a log of transactions
- Maintains before and after database images
- Maintains appropriate concurrency control
- All of the above.

XIII. A heterogeneous distributed database is which of the following?
- The same DBMS is used at each location and data are not distributed across all nodes.
- The same DBMS is used at each location and data are distributed across all nodes.
- A different DBMS is used at each location and data are not distributed across all nodes.
- A different DBMS is used at each location and data are distributed across all nodes.

XIV. Which of the following is not state that the transaction undergoes:
- Failed
- Active
- Partially committed
- Non active

XV. Transaction timestamp is:
- Not unique for each transaction
- Larger for older transaction
- Unique identifier assigned to each transaction.
- None of the above.

XVI. Assume transaction A holds a shared lock R. If transaction B also requests for a shared lock on R.
- It will result in a deadlock situation.
- It will immediately be granted.
- It will immediately be rejected.
- It will be granted as soon as it is released by A.

XVII. If a table has been normalized so that all determinants are candidate keys, then that table is in:
- 1NF.
- 2NF.
- 3NF.
- BCNF.

XVIII. Which type of entity is related to two or more associated entities that each contain specialized attributes that apply to some but not all of the instances of the entity?
- Supertype entity
- Subtype entity
- Archetype entity
- Instance entity

XIX. Which of the following is an SQL trigger supported by Oracle?
- BEFORE
- INSTEAD OF
- AFTER
- All of the above.

XX. Which of the following database activities determines the entities, attributes, and relationships of data?

- Conceptual data modeling
- Logical database design
- Physical database design
- Database implementation

Part - B

Attempt any four **4×2.5=10**

I. What are E-R diagrams? Explain the concepts of specialization and generalization.

II. What are the different kinds of DBMS users? Discuss the role of Database Administrator.

III. What is the motivation behind normalization? Discuss the first, Second and third normal with suitable example?

IV. Write difference between following:

- Where and Having Clause in SQL
- Stored Procedure and Stored Function
- DDL and DML SQL command
- Procedure and Trigger

V. What is transaction? Discuss different state of a transaction. Explain the properties of a transaction.

Part - C

Attempt any four **5×4=20**

I. What do you mean by serializability? Discuss the conflict and view serializability with suitable example.

II. Explain the concepts of Lock. What is two phase locking protocol? List the salient features of strict two phase locking protocol. Explain with suitable example how cascading rollbacks can be avoided using strict two phase locking.

III. What are deadlocks? Discuss different methods for dealing with deadlock problem?

IV. What is distributed database? Discuss the advantage of distributed database. Explain the data fragmentation with examples.

V. Why Concurrency Control is needed? Discuss basic timestamp ordering based concurrency control algorithm.

SET-B

B.Tech (CSE) II Year, IV Semester
End Semester Examination – 2012
Introduction to Microprocessor-CSE222

Time: 2 Hr 30 Mins **MM: 40**

Notes:-

1) Answer all questions, **Part A** is **compulsory**, **Any four** from **Part B** and **Any four** from **Part C**.
2) All parts of a question should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

Part A

Q1) Attempt following questions. **[0.5x20=10]**

i) Which provide a mechanism to establish a link between the microprocessor and I/O device?
- Input & Output both
- Output interface
- Input Interface
- None of these

ii) 8254 programmable timer counter has two inputs signals
- CLK & Gate
- Gate
- CLK
- None of these

iii) Which generate an interrupt to the microprocessor after a certain interval of time?
- 8259
- 8255
- 8251
- 8254

iv) The processor of knowing the status of device and transferring the data with matching speeds is called
- Ports
- Peripherals
- Handshaking
- None of theses

v) IRR stands for
- Input request register.
- Interrupt request register
- Interrupt resolver register.
- Input resolver register

vi) Which is widely used in interrupt controller with a number of microprocessor?
- 8255
- 8254
- 8251
- 8259

vii) In a 16-bit microprocessor, words are stored in two consecutive memory locations. The entire word can be read in one operation provided the first
- Memory is even
- word is odd
- word is even
- memory add is even

viii) In an 8085 based system, the maximum number of input output devices can be connected using I/0 mapped I/O method is
- 256
- 512
- 64
- 65536

ix) Interfacing devices for DMA controller, programmable interval timer are respectively
- 8257, 8251
- 8254, 8257
- 8257, 8253
- 8251, 8257

x) How many transistors does the 8086 have?
- 110,000
- 29,000
- 10,000
- 129,000

xi) The number of software interrupts in 8085 is
- 9
- 8
- 5
- 10

xii) In 8086 microprocessor one of the following instructions is executed before an arithmetic operation
- DAS
- AAD
- AAM
- DAA

xiii) In 8086 the over flow flag is set when
- Signed numbers go out of their range after an arithmetic operation.
- The sum is more than 16 bit
- Carry & Sign flag are set.
- Zero flag is set.

xiv) In 8085, example for non maskable interrupts is?
- INTR
- RST 6.5
- TRAP
- RST 5.5

xv) Which interrupts has highest Priority
- INTR
- TRAP
- RST 7.5
- RST6.5

xvi) In 8085 name the 16 bit registers.
- IR
- Program Counter
- SP
- PC & SP

xvii) Which of the following is a hardware interrupt?
- RST 5.5, RST 6.5, RST 7.5
- INTR, TRAP
- TRAP
- RST5.5,RST6.5,RST7.5,TRAP

xviii) What is Op-code?
- The value in which an operation acts upon
- The instruction that is to be executed
- A mnemonic that defines a data size
- The compiled assembly code

xix) What are the names of the 4 segment registers?
- Stack, Index, Extra, Code
- Data, Index, Code, Stack
- Stack, Data, Base, Counter
- Stack, Extra, Code, Data

xx) The 8086/8088 used two processing units which were known as
- Segment and Offset Units
- Left and Right Units
- Bus Unit and Execution Interface Unit
- Bus Interface Unit and Execution Unit

Part B

Q2) Attempt any four of the following. **[2.5x4=10]**

a) What is maximum memory addressing and I/O addressing capability of 8086?

b) List the internal registers of 8086 microprocessor with their abbreviations.

c) Write and explain all addressing modes of 8085 with examples?

d) Write a program to add the contents of the memory location 2000H:0500H to contents of 3000H:0600H and store the result in 5000H:0700H?

e) Explain the parallel I/O mode of 8255?

Part C

Q3) Attempt any four of the following. **[5x4=20]**

a) Draw the internal architecture of 8255 and mention the control word format of BSR Mode.

b) Explain the data transfer types supported by 8237 and also write the bit pattern of command register of 8237A?

c) Write and explain different mode of operation of 8253?

d) Discuss the organization of USART with a functional block diagram.

e) Discuss the following mode of operation of 8259

i) Cascade mode ii) Buffered mode iii) Nested mode

GLA University, Mathura

Set A

End Mid Term Examination, 2011-12

Course: - B.Tech (CS) II-Year, IV-Sem.

Subject: -Operating system **Subject Code:- CSE221** **Uni. Roll No:-**

Time:-150 Minutes **Total Marks:-40**

Notes:-

1) Answer all questions, Section A is **compulsory, Any Four** from Section B and **Any Four** from Section C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

SECTION-A

I Objective Questions **(10)**

I. The Dining Philosopher's Problem is a standard test case for evaluating approaches to implementing:

- Deadlock
- Starvation
- Synchronization
- Aging

II. A condition of policy that must be present for a deadlock to be possible is:

- Mutual exclusion
- Hold and wait
- No preemption
- Circular wait

III. LRU' page replacement policy is

- Last Replaced Unit.
- Last Restored Unit.
- Least Recently Used.
- Least Required Unit.

IV. The LRU algorithm

- Pages out pages that have been used recently
- pages out pages that have not been used recently
- Pages out pages that have been least used recently
- pages out the first page in a given area.

V. Thrashing

- is a Natural consequence of virtual memory systems
- can always be avoided by swapping
- always occures on large computers
- can be caused by poor paging algorithms.

VI. Interprocess communication

- is required for all processes
- is usually done via disk drives
- is never necessary
- allows processes to synchronize activity

VII. A system program that sets up an executable program in main memory ready for execution is

- assembler

- linker
- loader
- compiler.

VIII. The principal of locality of reference justifies the use of
- reenterable
- non reusable
- virtual memory
- cache memory.

IX. Fragmentation of the file system
- occurs only if the file system is used improperly
- can be always prevented
- can be temporarily removed by compaction
- is a characteristic of all file systems

X. To avoid the race condition, the number of processes that may be simultanelously inside their critical section is
- 8
- 1
- 10
- 0

XI. A Page fault
- is an error is a specific page
- occures when a program accesses a page of memory
- is an access to a page not currently in memory
- is a reference to a page belonging to another program.

XII. The dispatcher
- actually schedules the task into the processor
- puts tasks in I/o wait
- is always small and simple
- never changes task priorites

XIII. A disk scheduling algorithm in a operating system causes the disk arm to move back and forth across the disk surface in order to service all requests in its path. This is a
- FCFS
- SSTE
- SCAN
- FIFO

XIV. In which of the storage placement strategies a program is placed in the largest available hole in the main memory?
- best fit
- first fit
- worst fit
- buddy.

XV. What scheduling algorithm allows processes that are logical runnable to be temporarily suspended?
- Preemptive scheduling
- non-preemptive scheduling
- FIFO
- FCFS

XVI. Part of a Program where the shared moemory is accessed and which should be executed invisibly, is called
- semaphores

- directory
- critical section
- mutal exclusion.

XVII. Fork is a
- Creation of new job
- dispatching of a task
- increase a priority of a task
- the creation of a new process.

XVIII. A process can utilize a resources in the sequence
- Request,Use,Release
- Use, Request Release
- Use, Release,Request
- Request ,Release,Use

XIX. In order to allow only one process to enter its critical section, binary semphore are intitated to
- 0
- 1
- 2
- 6

XX. Solution of name collision problem is
- single Level directory
- Two level Directory
- Tree Structure
- Graph

SECTION-B

II Attempt any four **(10)**

1) List the various services provided by the operating system.
2) What do you understand by process? Explain various states of a process with suitalbe diagram.Explain Process Control Block.
3) Write a short note on:

a) Performance criteria for scheduling algorithms b) Multiprocessor Scheduling.

4) What is thrashing ? How does the system detect thrashing?
5) Explain the producer consumer problem. Give a solution to a problem using semphores.

SECTION-C

III Attempt any two **(20)**

1) Suppose the moving head disk with 200 tracks is currently serving a request for track 143 and has just finished a request for track 125. If the queue of requests is Kept in FIFO order: 86,147,91,177,94,150
   What is the total head movement for the following scheduling schemes:
   i) FCFS ii) SSTF iii) C-SCAN
2) What is dynamic linking and dynamic loading? Given memory partitions of 100 K, 500k , 200K, 300K and 600K(in order). How would each of the first-fit, Best fit and Worst –fit algorithms place processes of 212 K, 417K, 112K and 426K(In order)? Which algorithm makes the most efficient use of memory.
3) What are the functions of file system? List five system calls related to file system.
4) Distinguish between

i) Paging and segmentation

ii) Internal fragmentation and External Fragementation
iii) Logical address and Physical address
5) Consider the following page reference string:
1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6.
How many page faults would occur for the LRU,Optimal,FIFO algorithm using three frames only. Remember all frames are initially empty.

GLA University, Mathura
End Term Examination 2011-12
Course: B.Tech (CSE) II Year, IV semester

SET: C

Subject: Computer Graphics (CSE 223) **University Roll No.:**
Time: 2.30Hrs. **Total Marks: 40**
Notes:

1 Answer all questions. Part A is compulsory; answer any four from Part B and any four from Part C.
2 All parts of a question i.e. a,b,c,etc. should be answered at one place.
3 Answer should be brief, to the point and be supplemented with neat sketches.
4 Any missing or wrong data may be assumed suitably giving proper justification.
5 Figures at the right hand side margin indicate full marks.

Part A

(0.5x20=10)

Choose the most appropriate answer for the following multiple choice questions: for question(i) to (xx)

i The use of lookup tables in display monitor is to:
- *increase the resolution of the picture*
- *increase the range of colour shades*
- *store the colour information of the picture*
- *None of these*

ii In order to determine the visibility of the line against a rectangular window using endpoint codes, it is found that the codes for the line are 0101 and 0001. The line would be:
- *totally visible*
- *totally invisible*
- *partially visible*
- *lying along one of the edges of the window*

iii If a 1024 x 768 image is to be resized to one that is 640 pixels wide with the same aspect ratio, what would be the height of resized image in pixels.
- *256*
- *480*
- *512*
- *856*

iv Which format is used to store digital audio in multimedia application?
- *GIF*
- *WAV*
- *JPEG*
- *BMP*

v For which type of display the beam penetration method has not been used?
- *Calligraphic Display*
- *Random-scan Display*
- *Raster-scan Display*
- *Stroke-Writing Display*

vi A pixel from where we start filling is known as:
- *PEL*
- *ROOT*
- *SEED*
- *None of these*

vii Which one of the following transmits digital information to the computer?
- *joystick*
- *mouse*
- *graphics tablet*
- *None of these*

vii. The normal to any surface at any point is given by
- *cross product of parametric derivatives*
- *dot product of parametric derivatives*
- *scalar triple product of vectors*
- *None of these*

ix B-spline curves allow
- *Only the first order derivative to be continuous at the end points of curve*
- *Only the second derivative to be continuous at the end points of curve*
- *Both first and second derivative to be continuous at the end points*
- *First derivative at one end and the second derivative at the other end of the curve*

x Pick out the correct statement for Bezier curves.
- *Best curves are drawn when starting and end slope of the curves are same.*
- *A curve is proper only if the cubic polynomials for x and yare identical.*
- *The control points may lie anywhere on the screen.*
- *The control points should be so placed that values of x coordinate are in the increasing order.*

xi Which of the following are generated by successively dividing a two dimensional region into quadrants
- *Octree*
- *Binary space partitioning tree*
- *Quadtree*
- *None of these*

xii The maximum number of points that can be displayed without overlap on a CRT is referred as:
*Resolution *aspect ratio *pixel *None of these

xiii A many sided closed figure is known as:
*Polygon *Triangle *circle *Array

xiv The region of a display device into which a window is mapped, is called:
*Viewport *frame *box *image

xv Scanner uses:
*CCD array *LCD array *LED array *None of these

xvi Control points are used to control____ of curve:
*Shape *edges *values *iterations

xvii Total number. of objects that can be handled by z- buffer algorithm. is:
*fixed *arbitrary *only one *zero

xviii GKS is :
*Reliable *Portable *non portable *None of these

xix A display screen is:
*2D *3D *either2D or 3D *None of these

xx The distance between pixels on the screen is called:
*Dot pitch *OCR *Refresh rate *LCD

Part B (Attempt any four) **(2.5x4=10)**

1. (a) Describe the working of a CRT.
(b) What are the differences between the raster scan and random scan devices? (1.5+ 1)

2. (a) Describe the Bresenham's circle drawing algorithm.
(b) Explain how the Bresenham's line drawing algorithm works for the line joining the points (1, 2) and (7,5). (1.5+1)

3. (a) Describe the matrix forms of the two dimensional transformations of translation, rotation and scaling.
(b) Derive the transformation matrix for finding the reflection of a point with respect to the line $y = mx + c$. (1 + 1.5)

4. Perform a 45° rotation of triangle A(0, 0), B(1, 1), C(5, 2)
(a) about the origin and
(b) about point P(-1, -1) (1+ 1.5)

5. What are the various application areas of computer graphics? (2.5)

Part C (Attempt any four) **(5x4=20)**

1. Compare the various file formats used for image and video. Explain MIDI with its file format and compare it with digital audio.

2. Explain Back Face Detection algorithm. When there are concave polyhedral or overlapping objects, is Back Face detection sufficient? If yes, justify your answer. If no, suggest which techniques can be used and why?

3. What is compression? Write all steps for JPEG compression process in detail.

4. What are the components of a multimedia system? What are the hardware and software requirements of a multimedia computer?

5. What do you understand by virtual reality? Give the difference between multimedia applications and virtual reality applications with examples.

SET A

GLA University, Mathura

End Semester Examination, 2011-12

Course: - B.Tech. (CS) II-Year, IV-Sem.

Subject: -Computer networks **Subject code-CSE-224** **Uni. Roll No:-**

Time: - 150 Minutes **Max. Marks:-40**

Notes:-

1) Answer all questions, Part A is **compulsory**, **Any Four** from Part B and **Any Three** from Part C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
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Part-A (Attempt all questions) [1/2*20]

i. **Which error detection method uses ones complement arithmetic?**

- CRC
- Simple Parity Check
- Hamming Code
- Check-Sum

ii. **Tunneling is a concept used for——.**

- Fragmentation
- error control
- transition from IPV4 to IPV6
- multicast routing

iii. **In Go-Back-N ARQ, if frames 4, 5, and 6 are received successfully, the receiver may send an ACK _____ to the sender.**

- 5
- 7
- 6
- Any of the these

iv. **In fiber optics, the signal source is _____ waves.**

- light
- radio
- micro
- very low frequency

v. **The data unit is 111101, the divisor 1101, what is the dividend at the transmitter?**

- 111111000
- 001
- 10000
- None of these

vi. **What is the default subnet mask for a class C network?**

- 127.0.0.1
- 255.255.0.0
- 255.0.0.0
- 255.255.255.0

vii. **The Internet Control Message Protocol (ICMP) is**

a TCP/IP protocol used to dynamically bind a high level IP Address to a low-level physical hardware address

a TCP/IP high level protocol for transferring files from one machine to another

- a protocol used to monitor computers
- a protocol that handles error and control messages

viii. **Compared to pure ALOHA, slotted ALOHA has:**

- higher efficiency, because without slots a frame is more likely to suffer a collision.
- higher efficiency, because slotted Aloha uses collision avoidance.
- lower efficiency, because a large part of a slot can go unused.
- lower efficiency, because having slots requires synchronization.

ix. **Which of the following TCP/IP protocol is used for transferring electronic mail messages from one machine to another?**

- FTP
- SNMP
- SMTP
- RPC

x. **These can impair a signal on a network.**

- noise
- distortion
- attenuation
- all of these

xi. **An organization is granted the block 130.34.12.64/26. The organization needs to have four subnets. What is the subnet mask?**

- 255.255.255.0
- 255.255.255.192
- 255.255.255.240
- None of these

xii. **A network on the Internet has a subnet mask of 255.255.240.0. What is the maximum number of hosts it can handle?**

- 1024
- 2048
- 4094
- None of these

xiii. **Suppose that instead of using 16 bits for the network part of a class B address originally, 20 bits had been used. How many class B networks would there have been.**

- 2^{20}.
- 2^{20} -2.
- 2^{18}-2
- None of these

xiv. **What is the minimum size of the TCP header?**

- 40 byte.
- 20 byte
- 60 byte
- 80 byte

xv. **An IPV4 packet has arrived in which the offset value is 100. What is the number of the first byte?**

- 400
- 800
- 100
- 200

xvi. **In the CSMA/CD protocol, what condition on the transmission delay T_{trans} and the propagation delay T_{prop} has to be satisfied to guarantee that a node always detects a collision?**

- $T_{trans} > T_{prop}$
- $T_{trans} > 2\ T_{prop}$
- $T_{trans} < T_{prop}$
- $2\ T_{trans} < T_{prop}$

xvii. **What is the size of the window for host A, if the value of receiver window (rwnd) is 3000 bytes and the value of congestion window (cwnd) is 3500 bytes?**

- 3000 byte
- 3500 byte.
- 500 byte
- 1000 byte.

xviii. **Host A sends a TCP segment (Seq = 43), to which host B replies with a TCP segment (ACK = 57). The size of TCP segment is ———.**

- 14 bytes long.
- 46 bytes long.
- 43 bytes long.
- 57 bytes long.

xix. **Which of the following is incorrect about IPv6?**

- The size of IPv6 addresses is 128 bits.
- There is no checksum in the IPv6 header.
- There is fragmentation offset in IPV6 header.
- There is no HLEN field in IPV6 header

xx. **DHCP:**

- allows an ISP to obtain a set of IP addresses from the ICANN.
- allows a router to allocate port numbers in a NAT (Network Address Translation).
- allows a host to dynamically obtain an IP address when it joins the network.
- Maps logical address into physical address.

Part-B (Attempt any Four) **[2.5x4=10]**

I. Assume that a router has the following entries in its routing table:

Address/mask	Next hop
135.46.56.0/22	Interface 0
135.46.60.0/22	Interface 1
192.53.40.0/23	Interface 2
default	Interface 3

For each of the following IP addresses, what does the router do if a packet with that destination address arrives? :

a) 135.46.63.10
b) 135.46.57.14
c) 135.46.52.2
d) 192.53.40.7
e) 192.53.56.7

II.

a) Explain the leaky bucket algorithm?
b) What is the serious drawback of distance vector routing? Explain it with example.

III. Explain Link State Routing algorithm.

IV.

a) Describe the characteristics and application of Bridge.

b) Draw the graph of manchester and differential manchester scheme for the following data stream-11011011

V. Consider an error-free 64-kbps satellite channel used to send 512-byte data frames in one direction, with very short acknowledgements coming back the other way. What is the maximum throughput for window sizes of 1, 15, and 127? The earth-satellite propagation time is 270 msec.

Part-C (Attempt any Three) **[20]**

I.

a) Explain TCP retransmission time-out (RTO) and fast retransmission techniques for error control.

b) How does DNS work? Explain with example.

II. What is congestion? Explain TCP slow start and congestion avoidance algorithms with example.

III.

a) Using the RSA public key cryptosystem, with a = 1, b = 2, etc., Using p = 5, q = 11, and d = 27, find e and encrypt "**ab**".

b) Change the multicast IP address 238.212.27.19 to an Ethernet multicast address.

IV. Write short notes on **any Two-**

a) ARP b) FTP c) POP3